

# Executable Integrations on POCKET: Desktop-First Connectors, Browser Fallbacks, and Intentional Life Ops

## Working Paper

*System:* POCKET Integrations Execute

*Schema:* `pocket.integrations.execute.v1`

*Lab:* ItsNotAI Labs

*Product:* POCKET

*Date:* 2026-08-08

*Status:* First-class — 55 catalog entries, live execute path

## Abstract

Integration catalogs that only open marketing URLs are **directory UX**, not agency. This paper specifies and evaluates **POCKET Integrations Execute**: a uniform execute surface where every catalog entry can produce a **structured receipt** via desktop app launch, signed-in Edge remote browser, Working board intent, agent seat, or host tab.

On the operator Windows host we measure **55/55 dry-run execute**, desktop resolution for Discord (latest app-\*), Teams, Edge, GitHub Desktop, and Copilot, and board+browser paths for dining/commerce/productivity SaaS. We document the safety allowlist, Discord Squirrel/update.exe handling, and the doctrine that **execute is first-class**—list is not enough.

*Keywords:* desktop automation, connectors, OAuth-less signed-in browser, integration catalog, POCKET, host agents

## 1. Introduction

Agent products frequently list “50+ integrations” that merely deep-link to websites. Users discover that “Discord integration” means a tab to `discord.com/app`, not launching their installed client with their real sessions.

POCKET’s requirement is different: **agents and the desk must be able to act**. This paper defines the execute contract, maps actions to host surfaces, and reports live host results.

### 1.1 Contributions

- 1 Unified **execute** engine (`integrations_exec.execute`) for all catalog entries

- 2 **Desktop-first** mapping for installed apps with browser fallback
- 3 **Board intents** for reserve/buy/research/notify without auto-pay
- 4 Live operator-host evaluation (55-entry catalog)
- 5 Wiring to desk UI, MCP skills, and LOOMGRAPH integrate nodes

## 2. Design

### 2.1 Catalog model

Each integration:

```
id, name, category, icon, blurb, action, url?, desktop_app?, prefer?, agent?, tab?, prompt?, featured?, executable?
```

**Actions:** open | reserve | buy | research | notify | schedule | errand | analysis | agent | remote |  
screen | tab | working

### 2.2 Execute modes

| Mode    | Behavior                                              |
|---------|-------------------------------------------------------|
| Desktop | desktop.open_app under safety allowlist               |
| Browser | remote_browser.open_url (signed-in Edge profile)      |
| Board   | Start Working board intent + optional browser/desktop |
| Agent   | Seat agent + optional browser warm                    |
| Host    | Screen control / MCP tab / working board              |

### 2.3 Discord special case

Discord on Windows often resolves to update.exe. POCKET:

- 1 Prefers newest %LOCALAPPDATA%\Discord\app-\*\Discord.exe
- 2 Falls back to update.exe --processStart Discord.exe
- 3 Else opens https://discord.com/app in Edge

### 2.4 Safety

- Only safety.ALLOWED\_APPS keys may launch
- Audit log on open
- No arbitrary executable paths from the agent

### 2.5 API

```
GET /v1/integrations
GET /v1/integrations/readiness
POST /v1/integrations/execute
  {"id":"discord","prefer":"desktop"}
POST /v1/integrations/execute_all
```

```
{"dry_run":true}
```

Skills: integrations\_list, integrations\_execute, integrations\_readiness

### 3. Catalog composition (v1)

*Count: 55 executable entries*

Categories include AI, Dining, Productivity, Dev, Cloud, Commerce, Social, Media, Design, Files, Meetings, CRM, Automation, Host.

*Desktop-mapped examples: Discord, Slack, Teams, Spotify, Zoom, Notion, Figma, GitHub Desktop, Outlook, Linear, Edge, Telegram, ChatGPT, Copilot, Obsidian*

*Featured: Muse Spark, OpenTable, GitHub, Cloudflare, Amazon, Edge, Screen Control, MCP, Working board, Discord, Copilot*

### 4. Evaluation

#### 4.1 Dry-run matrix

execute\_all(dry\_run=True) → **55/55 ok** Every integration produces a non-empty step plan (desktop planned and/or browser planned and/or host ready).

#### 4.2 Readiness (operator host)

| Metric                        | Value                                              |
|-------------------------------|----------------------------------------------------|
| Catalog count                 | 55                                                 |
| Browser URL surfaces          | 50                                                 |
| Desktop available (installed) | 5+ (Copilot, Discord, GitHub Desktop, Teams, Edge) |
| Board intents                 | 15                                                 |

#### 4.3 Live execute samples

| id             | Result                     |
|----------------|----------------------------|
| discord        | desktop:done → Discord.exe |
| teams          | desktop:done               |
| browser_edge   | remote_browser:done        |
| github         | pick_agent:ready           |
| working_board  | working_board:ready        |
| screen_control | screen_control:ready       |
| mcp_tools      | show_tab:ready             |

| id        | Result                                    |
|-----------|-------------------------------------------|
| spotify   | desktop unavailable → browser:done        |
| notion    | desktop unavailable → browser:done        |
| opentable | working_board + browser:done              |
| gmail     | working_board + browser:done              |
| slack     | board path; browser warm when URL present |

## 4.4 Desk UX

useIntegration(id) POSTs execute first, then seeds prompts / Working board / tabs. Failures toast with receipt messages.

## 5. Discussion

### 5.1 “Working” definition

An integration is **working** if execute returns ok=true with at least one of:

- 1 Desktop process started
- 2 Browser navigation started
- 3 Host surface ready (board/tab/screen/agent)

Missing install is **not** a catalog failure if browser/board fallback succeeds.

### 5.2 Why not OAuth for everything

Host Edge already carries user sessions for Gmail, Notion, dashboards. Desktop apps carry native sessions. OAuth multiplies key management for operators; POCKET prefers **session reuse** with explicit safety allowlists. Enterprise OAuth bridges remain future work.

### 5.3 Composition with LOOMGRAPH

LOOMGRAPH graph integration\_open calls integrations\_exec.execute as an integrate node, so Discord open appears as a path segment humans can see:

sense → list → exec → verify → done

## 6. Threats and mitigations

| Threat               | Mitigation                                          |
|----------------------|-----------------------------------------------------|
| Arbitrary exe launch | Allowlist only                                      |
| Silent public share  | Community is separate; integrations never auto-post |
| Auto-purchase        | Board intents require human confirm; never auto-pay |
| Stale Discord path   | Dynamic app- * discovery                            |

## 7. Conclusion

POCKET Integrations Execute turns a connector directory into **agency**. With 55 executable entries, desktop-first Discord/Teams, Edge fallbacks, and board intents for life ops, the host can honestly claim **working integrations**—backed by receipts, not marketing URLs alone.

## Appendix A — Live smoke (2026-08-08)

```
count 55 desktop 5 browser 50
dry_all 55 / 55 fail 0
discord          ok=True desktop:done
browser_edge     ok=True remote_browser:done
teams           ok=True desktop:done
github          ok=True pick_agent:ready
working_board    ok=True working_board:ready
```

## Appendix B — Citation

ItsNotAI Labs (2026). *Executable Integrations on POCKET*. Working Paper. Schema  
`pocket.integrations.execute.v1`.

*End of working paper.*